SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 9

1. Define a hash table and explain how it stores key-value pairs.
2. Describe the role of a hash function in a hash table.
3. Explain a collision in a hash table and the reasons it occurs.
4. Outline how separate chaining resolves collisions in a hash table.
5. Discuss the difference between open addressing and separate chaining for collision resolution.
6. Illustrate how linear probing handles collisions in a hash table.
7. Given a hash table of size 5 and the hash function `hash(key) = key % 5`, insert the following keys: 12, 7, 8, and 4.
   i) Show the placement of the keys in the hash table.
   ii) Discuss the outcome when inserting the key 2 into the hash table.
8. Given a hash table of size 7 and the hash function `hash(key) = key % 7`, insert the following keys: 19, 25, 31, and 3.
   i) Show the placement of the keys in the hash table using linear probing.
   ii) Show the placement of the keys in the hash table using separate chaining.